Terraform Notes (Quick Guide)

◈ What is Terraform?

- **Terraform** is an open-source tool by HashiCorp
- Used to **create, manage, and automate infrastructure** (servers, VPC, EC2, etc.)

☞ Example: Instead of manually creating AWS resources, you write code.

◈ Infrastructure as Code (IaC)

- Writing infrastructure using code
- Version control (Git)
- Reusable & automated

◈ Terraform File Types

- `.tf` → Main configuration files
- `variables.tf` → Input variables
- `outputs.tf` → Output values
- `terraform.tfvars` → Variable values

◈ Basic Terraform Workflow

1. **terraform init**
   ☞ Initialize project
2. **terraform plan**
   ☞ Preview changes
3. **terraform apply**
   ☞ Create resources
4. **terraform destroy**
   ☞ Delete resources

◈ Basic Example (AWS EC2)

```
provider "aws" {
  region = "us-east-1"
}

resource "aws_instance" "example" {
  ami           = "ami-12345678"
  instance_type = "t2.micro"
}
```

◈ Key Concepts

1. Provider

- Platform where resources are created
  ☞ AWS, Azure, GCP

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2. Resource

- Actual infrastructure
  ☞ EC2, VPC, S3

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3. Variables

```
variable "region" {
  default = "us-east-1"
}
```

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4. Output

```
output "instance_ip" {
  value = aws_instance.example.public_ip
}
```

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5. State File

- `terraform.tfstate`
- Stores current infrastructure info

⚠ Important: Do not delete or edit manually

◈ Terraform Commands (1-Line)

- `terraform init` → Setup Terraform
- `terraform plan` → Show changes
- `terraform apply` → Execute changes
- `terraform destroy` → Remove infra
- `terraform validate` → Check syntax
- `terraform fmt` → Format code

◈ Advantages

✅ Automation
✅ Easy to reuse
✅ Multi-cloud support
✅ Version control

◈ Disadvantages

❌ State file management
❌ Learning curve
❌ Errors can affect infra

◈ Interview Questions

1. What is Terraform?

👉 Tool to manage infrastructure using code.

2. What is state file?

👉 Stores current infra details.

3. Difference between plan & apply?

👉 Plan = preview, Apply = execution.

4. What is provider?

👉 Platform (AWS/Azure/GCP).

5. What is resource?

👉 Infrastructure object (EC2, VPC).

◈ Real-Time Use Case

👉 CI/CD pipeline:

- Developer pushes code (GitHub)
- Jenkins runs Terraform
- Infrastructure created automatically